

Saeed Talamkhani

PhD Candidate
Department of Civil, Environmental, and Ocean Engineering
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EDUCATION

- May 2027 PhD, Civil Engineering, Stevens Institute of Technology, Hoboken, NJ
(expected) Dissertation: *Advanced underwater vision-based robotics for automated infrastructure inspection and maintenance*
- 2018 MS, Civil Engineering, Imam Khomeini International University, Qazvin, Iran
Thesis: *Effect of Geotextile on the Undrained Shear Strength of Clayey Sand*
- 2015 BS, Civil Engineering, University of Qom, Qom, Iran

RESEARCH AREAS

Civil infrastructure systems	Artificial intelligence
Underwater enhanced perception	Computer vision
Automated task planning	Deep learning
Robotic systems	Semantic modeling

PUBLICATIONS

Journal Articles

- 2026 **S. Talamkhani** and K. Liu. "Knowledge-guided Few-shot Learning Segmentation of Multi-class Damages for Supporting Underwater Bridge Components from Limited Field Inspection Images." *Automation in Construction*, (to be submitted).
- 2026 J. Xu, H. Yu, **S. Talamkhani**, Y. Ye, A. Hedayat, K. Liu, and Y. Shi. "Virtual and Augmented Reality in Underground Space: a Systematic Review of 173 Studies." *Computing in Civil Engineering*, (to be submitted).
- 2025 **S. Talamkhani** and K. Liu. "Underwater vision-enhanced image segmentation for supporting automated inspection of underwater bridge components." *Automation in Construction*, 175 (2025): 106230. doi:10.1016/j.autcon.2025.106230
- 2023 **S. Talamkhani**. "Machine Learning-Based Prediction of Unconfined Compressive Strength of Sands Treated by Microbially-Induced Calcite Precipitation (MICP): A Gradient Boosting Approach and Correlation Analysis." *Advances in Civil Engineering*, 1 (2023): 3692090. doi:10.1155/2023/3692090
- 2023 **S. Talamkhani**, S.A. Naeini and A. Ardakani. "Prediction of Static Liquefaction Susceptibility of Sands Containing Plastic Fines Using Machine Learning Techniques." *Geotechnical and Geological Engineering*, 41, (2023): 3057–3074. doi:10.1007/s10706-023-02444-2
- 2021 **S. Talamkhani** and S.A. Naeini. "The Undrained Shear Behavior of Reinforced Clayey Sand." *Geotechnical and Geological Engineering*, 39, (2021): 265–283. doi:10.1007/s10706-020-01490-4
- 2018 **S. Talamkhani** and S.A. Naeini. "Effect of Plastic Fines on Undrained Behavior of Clayey Sands." *International Journal of Earth, Energy and Environmental Sciences*, 11, 0, (2018): 525–528. doi:10.5281/zenodo.1340402

Conference Proceedings

- 2026 **S. Talamkhani** and K. Liu. "Structured Pruning-Based Efficiency Optimization of Deep Vision Models for Automated Infrastructure Inspection." *Proceedings of the 2026 International Conference on Industrialized Construction (IConIC)*. Gainesville, FL: University of Florida. (In Press).
- 2025 **S. Talamkhani**, Y. Shi, and K. Liu. "An Ontology Approach to Context-Aware Robotics for Automated Underwater Bridge Maintenance." *Proceedings of the 2025 ASCE International Conference on Computing in Civil Engineering (i3CE)*. New Orleans, LA: Louisiana State University. (2025):679-688. doi:10.1061/9780784486443.07
- 2024 **S. Talamkhani** and K. Liu. "Integrating Image Enhancement, Classification, and Segmentation for Supporting Automated Underwater Infrastructure Inspection and Maintenance." *Proceedings of the 2024 ASCE International Conference on Computing in Civil Engineering (i3CE)*. Pittsburgh, PA: Carnegie Mellon University. (2024):229-238. doi:10.1061/9780784486443.074

POSTER PRESENTATIONS

- 2025 "Advanced Underwater Robotic System for Automated Underwater Infrastructure Inspection and Maintenance." *AI in Construction Workshop*. Hoboken, NJ: Stevens Institute of Technology.
- 2024 "Advanced Underwater Vision for Automated Underwater Infrastructure Inspection and Maintenance." *Proceedings of the 2024 ASCE International Conference on Computing in Civil Engineering (i3CE)*. Pittsburgh, PA: Carnegie Mellon University.

HONORS AND AWARDS

- 2026 Runner-up Paper Award in IConIC 2026
- 2023 The Provost Doctoral Fellowship

TEACHING EXPERIENCE

- 2025 F CM 506 – Computer Applications in the Construction Process, Stevens Institute of Technology
- Led seven hands-on sessions and instructed state-of-the-art computer applications and workflows in construction using BIM-based software tools: *Revit* for design development and construction co-ordination; *Navisworks* for clash detection, cost estimation, and scheduling; *Solibri* for compliance checking; *Infraworks* for civil information modeling
 - led two sessions of case study and final project presentation
 - Held weekly office hour to help students on their assignments problems
- 2025 S CE 483 – Geotechnical Engineering, Stevens Institute of Technology
- Attended ten lab-sessions and assisted the instructor to help students performing geotechnical tests
 - Graded five of lab reports for five geotechnical test
 - Graded two midterm exams and one final exams
 - Held weekly office hour to help students on their assignments problems
- 2024 F CE 365 – Numerical Modeling in Civil and Environmental Engineering, Stevens Institute of Technology
- Instructed *numerical differentiation* and *numerical integration* two sessions during the semester
 - Led two mid-term and one final exams
 - Held weekly office hour to help students on their assignments problems
 - Graded weekly MATLAB coding assignments
 - Graded two midterm exams and one final exams
 - Created Canvas–SpeedGrader rubrics that sped up grading and made exam scoring transparent with point-by-point feedback students could review online

ACADEMIC SERVICE

Journal Peer Review

Journal of computing in civil engineering, ASCE

Environmental science and pollution research international, Springer

Bulletin of engineering geology and the environment, Springer

Conference Proceeding Peer Review

Proceedings of the 2026 ASCE International Construction Research Congress (CRC)

Proceedings of the 2026 ASCE International Conference on Computing in Civil Engineering (i3CE)

Proceedings of the 2025 ASCE International Conference on Computing in Civil Engineering (i3CE)

MEMBERSHIPS

Jul 2024 American Society of Civil Engineers (ASCE)

Jul 2024 Visualization, Information Modeling and Simulation (VIMS) Committee, ASCE

Nov2023 Data Sensing and Analysis (DSA) Committee, ASCE

Updated October 2026